PROJECT PLAN AND GANTT CHART

PROJECT NAME	Advisor	START DATE	END DATE
Smart Migraine Monitoring Device	Dr. Fady Rouphael	1-sep	16-dec

OVERALL PROGRESS
100%

PROJECT DELIVERABLE
(Chapter 4 - Chapter 5 - Chapter 6- Chapter 7 – Chapter 8 - Final report - Power point presentation- Gantt Chart)

SCOPE STATEMENT
A safe, effective, and user-friendly smart migraine monitoring device that supports early migraine detection and prevention.

	Task	Assigned To	Start	End	Dur
	Project Management		9/1/25	12/19/25	80
1	Project Planning & Scope Definition	Lana, Rafic and Nicolas	9/1/25	9/10/25	7
2	Medical Background	Lana and Nicolas	9/1/25	9/5/25	5
3	Project Requirements & Specifications	Lana, Rafic and Nicolas	9/1/25	9/8/25	6
4	ESP32 Power System Implementation	Nicolas and Lana	9/12/25	9/16/25	3
5	Sensor Integration (MAX30102, HTU21D)	Nicolas and Rafic	9/18/25	9/25/25	6
6	Battery Charging & Regulation Testing	Nicolas	10/2/25	10/7/25	4
7	Vibration Motor & MOSFET Integration	Nicolas	10/5/25	10/9/25	4
8	Hardware Validation & Debugging	Lana, Rafic and Nicolas	12/1/25	12/13/25	10
9	ESP32 Firmware Development	Rafic	10/5/25	10/10/25	5
10	Sensor Data Processing & Filtering	Rafic and Nicolas	10/3/25	10/16/25	10
11	Firebase Integration (RTDB + Streaming)	Lana and Nicolas	9/15/25	11/11/25	42
12	ESP32 Software Testing	Nicolas	11/20/25	11/29/25	7
13	Battery Monitoring & Alerts Logic	Rafic	11/27/25	11/30/25	2
14	Firebase ↔ App Integration	Lana and Nicolas	9/15/25	11/5/25	38
15	Data Visualization & Alerts Display	Rafic	12/1/25	12/3/25	3
16	App Functional Testing	Nicolas and Rafic	11/15/25	12/1/25	11
17	End-to-End Data Flow Testing	Rafic	12/10/25	12/13/25	3
18	Chapter 4 (Design & Updates)	Lana and Nicolas	11/20/25	11/26/25	5
19	Chapter 5 (Procedures)	Lana and Rafic	12/10/25	12/15/25	4
20	Chapter 6 (Software)	Lana	12/10/25	12/12/25	3
21	Chapter 7 (Hardware)	Lana, Rafic and Nicolas	11/25/25	12/1/25	5
22	Final Editing & Formatting	Lana	12/15/25	12/18/25	4
23	Presentation Preparation	Lana, Rafic and Nicolas	12/1/25	12/15/25	11
24	Oral Defense	Lana, Rafic and Nicolas	12/19/25	12/19/25	1